



Exploratory Data Analysis

From cleaned data to evidence-based questions

Lecture 7

ITDS341 Fundamentals of Data Science

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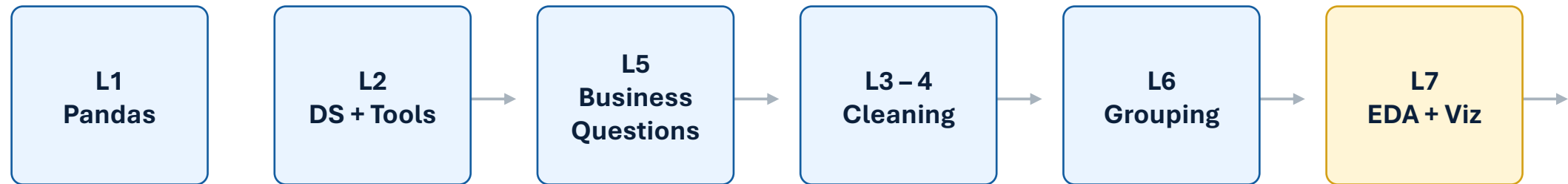
Learning Outcomes

- Select summaries and plots based on feature types.
- Explore one variable and describe shape, center, spread, gaps, and unusual values.
- Compare variables and groups using suitable visualizations and summaries.
- Interpret relationships cautiously: association is not causation.
- Turn plots into observations, insights, and follow-up questions.

Agenda

- Exploratory Data Analysis
- Feature Types
- Univariate Data Analysis
- Multivariate Data Analysis
- Hypothesis / Insight

Learning Path ... so far



**The goal today: use the data you have cleaned and summarized
to discover patterns worth investigating.**



Exploratory Data Analysis

Exploratory Data Analysis

*Exploratory data analysis is actively incisive, rather than passively descriptive, with real emphasis on the **discovery of the unexpected**.*

- **Goals:** Guide our quality checks of our data cleaning and transformations.
- Common techniques:
 - Summary statistics
 - Visualization

Exploratory Data Analysis

- In EDA, we plots to uncover **features** of the data, examine **distributions** of values, and reveal **relationships** that **CANNOT** be detected from simple numerical summaries.
- Best ways to learn EDA: **Learn from others** as they describe their thought process while they explore data.
- EDA can provide valuable insights, **but** you need to be cautious about the conclusions that you draw.
- It is important to recognize that EDA can **bias your analysis**.

Exploratory Analysis

*The greatest gains from data come from **surprises...** The unexpected is best brought to our attention by **pictures**.*

- To make these pictures, we need to **choose an appropriate type of plot**, and our choice **depends on the kinds of data** that have been collected.

EDA Is Not a Plot Gallery

Weak approach
“Make every chart I know.”



Better approach
Ask → inspect → visualize → interpret → ask again

EDA is a thinking process.

A good plot should lead to a **clearer observation** or a **better next question**.

From Data Preparation to EDA



Cleaning and exploration are iterative.

EDA can expose missingness, strange distributions, outliers, imbalance, or bias that were not obvious at first.

Start With an EDA Question

Distribution

How is X distributed?

Comparison

How does X differ across groups?

Relationship

How do X and Y move together?

Quality check

Are there missing, invalid, or unusual values?

A question decides what evidence you need. The evidence decides what plot or summary is useful.



Feature Types

Feature Types (Recap)

- **Numerical Continuous:**
Measurements that can take any value within a continuous range.
 - BMI, charges
- **Numerical Discrete:**
Measurements of counts that take separate, countable values, usually whole numbers.
 - number of children, step count
- **Categorical Nominal:** A feature that represents “named” categories, where the categories do not have a natural ordering.
 - gender, smoker, region
- **Categorical Ordinal:** Measurements that represent ordered categories.
 - low / medium / high
 - Likert-scale response (disagree, neutral, agree)

dtype ≠ analytical meaning

- `df.dtypes` → storage type
- `df.nunique()` → number of unique values
- `df['sex'].value_counts()` → observed categories/frequencies

Determine feature type from what the variable means — not dtype alone.

For example, a column can have dtype `int64` but still be categorical:

```
sex = 0, 1
```

Transforming Features for Exploration

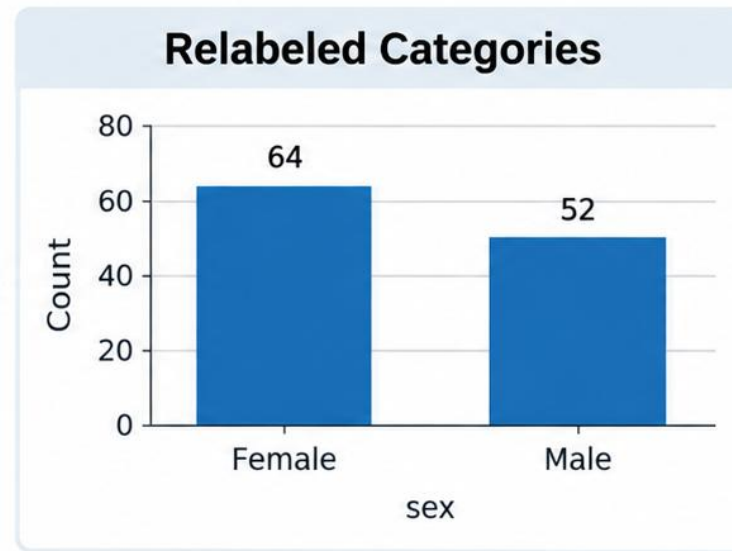
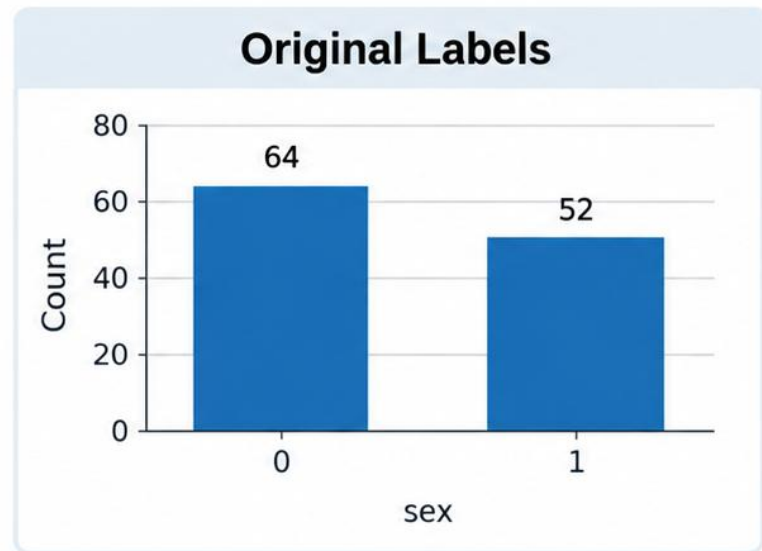
Sometimes we transform a feature to make patterns easier to interpret.

- **Relabel Categories:** Relabel categories so that they are more informative (0 $\rightarrow$ female, 'f' $\rightarrow$ female)
- **Collapse Categories:** A categorical feature has many categories and we prefer a higher level view of the data so we collapse categories.
- **Convert Quantitative to Ordinal:** When a discrete quantitative feature has a long tail, that we want to truncate the higher values, which turns the quantitative feature into ordinal.

Transforming Categorical Features

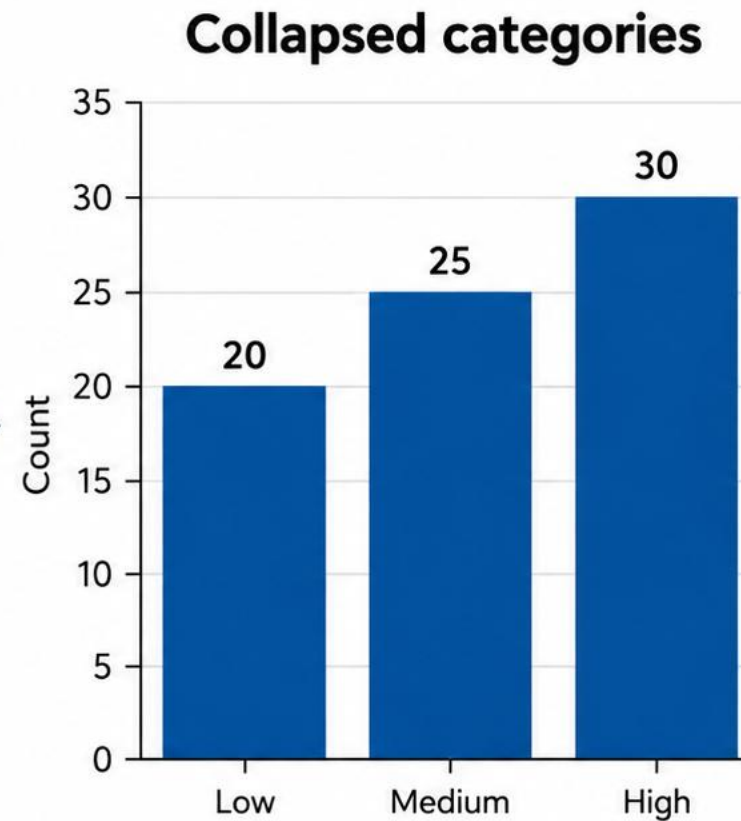
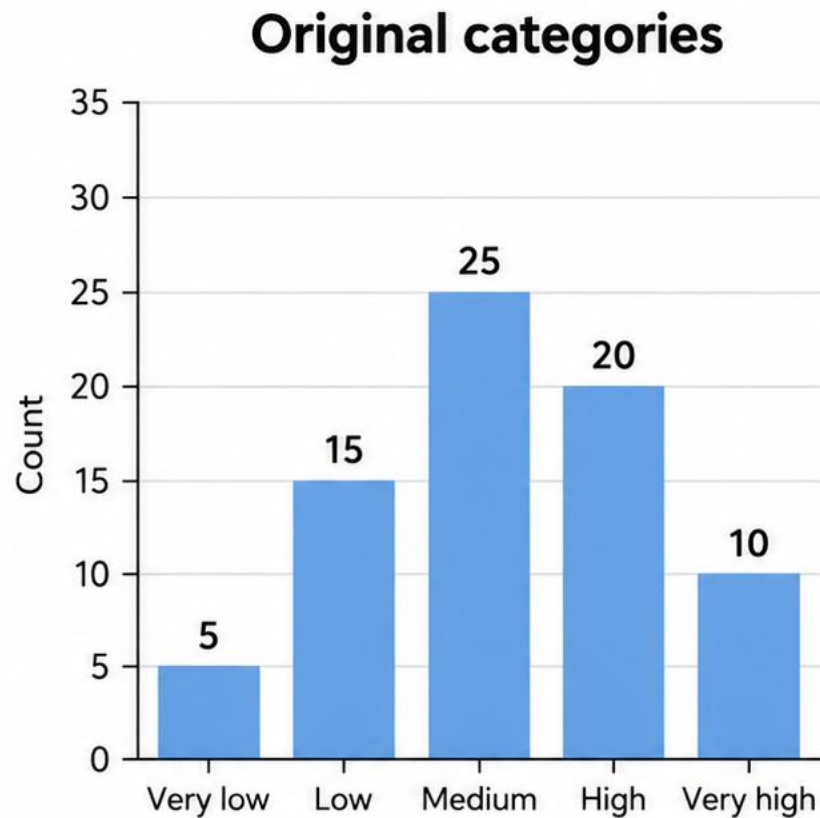
Sometimes we transform a feature to make patterns easier to interpret.

- **Relabel Categories:** Relabel categories so that they are more informative (0 → female, 'f' → female)



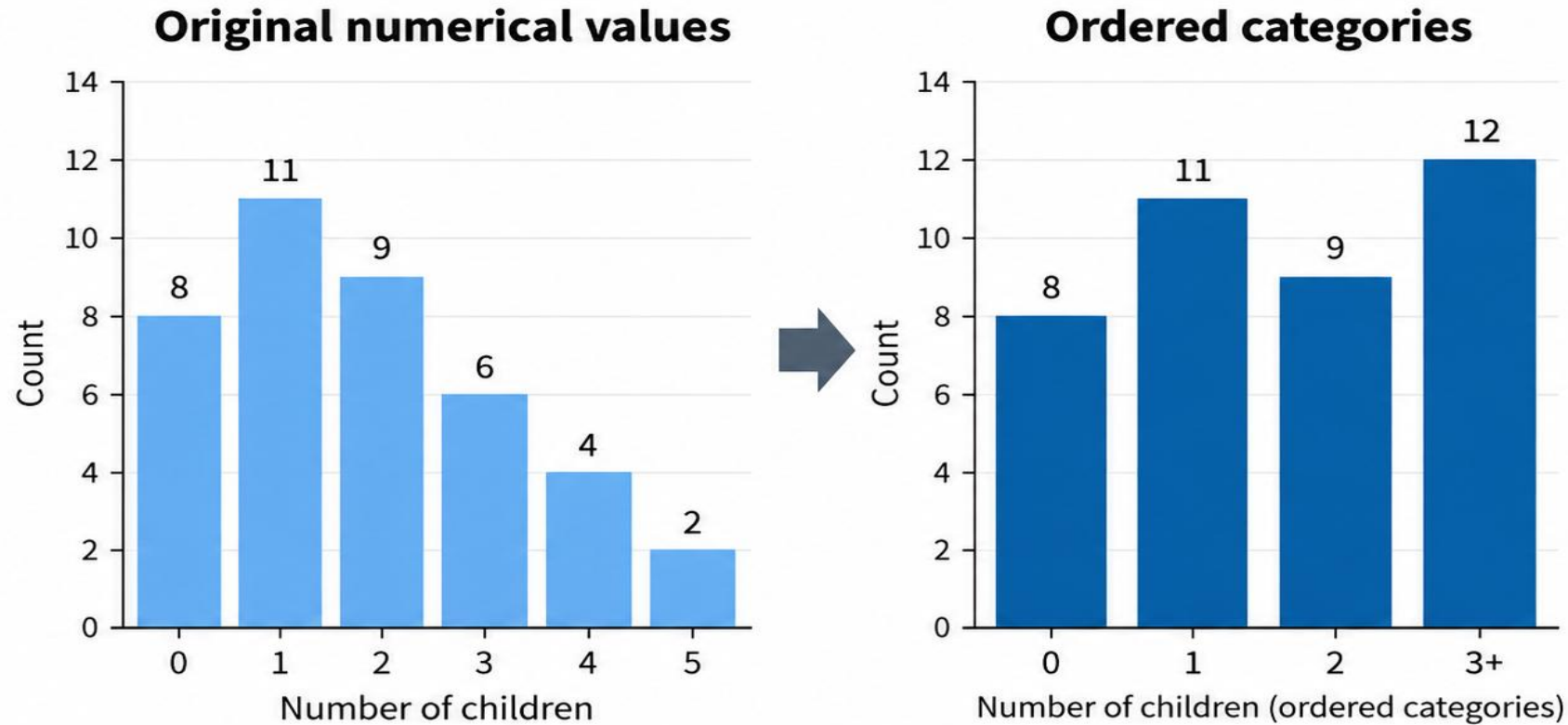
Change from coded labels (0, 1) or short labels (F, M) to **meaningful labels** (Female, Male) to make the data easier to interpret.

- **Collapse Categories:** A categorical feature has many categories and we prefer a higher level view of the data so we collapse categories.

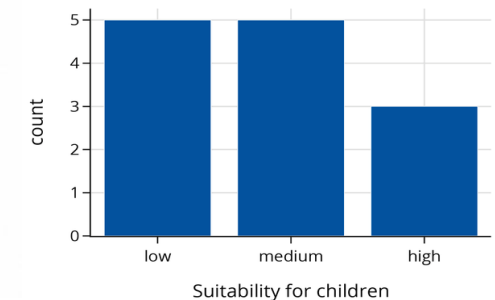


Change from detailed categories (**very low, low, medium, high, very high**) to broader groups (**low, medium, high**) to make the overall pattern easier to interpret.

- **Convert Quantitative to Ordinal:** When a discrete quantitative feature has a long tail, that we want to truncate the higher values, which turns the quantitative feature into ordinal.



Change from a discrete numerical variable (0, 1, 2, 3, 4, ...) to **ordered categories** (0, 1, 2, 3+) to make grouped comparisons easier.



Note: This simplifies interpretation, but loses some detail.

Choose a View Based on Question + Feature Type

Variables	Question	Starting evidence
One categorical	How common are categories?	value_counts + bar chart
One numerical	What is the shape?	describe + histogram / box plot
Two numerical	Do they move together?	scatterplot + correlation
Numerical + categorical	How does X differ by group?	groupby + box plot
Two categorical	Do proportions differ?	crosstab + grouped bar

Step 0: Quick Data Readiness Check

```
df.shape  
df.info()  
df.isna().sum()  
df.describe(include='all')
```

Before visualizing, quickly check whether the data is ready enough to explore.

Rows and columns

Data types

Missing values

Basic ranges and categories

Only fix what blocks exploration.

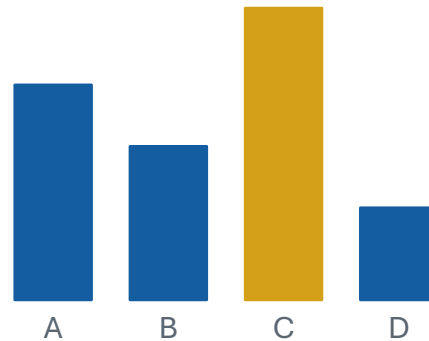


Univariate Data Analysis

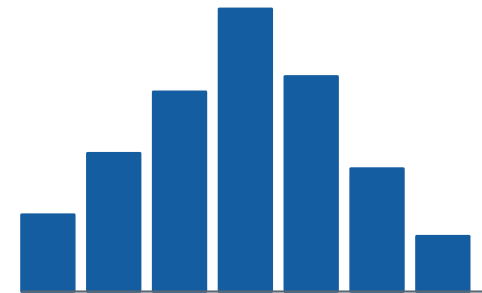
Univariate Data Analysis

- Descriptive Summary
- Categorical variable
 - Frequency table
 - Frequency chart
- Continuous variable
 - Histogram
 - Box plot

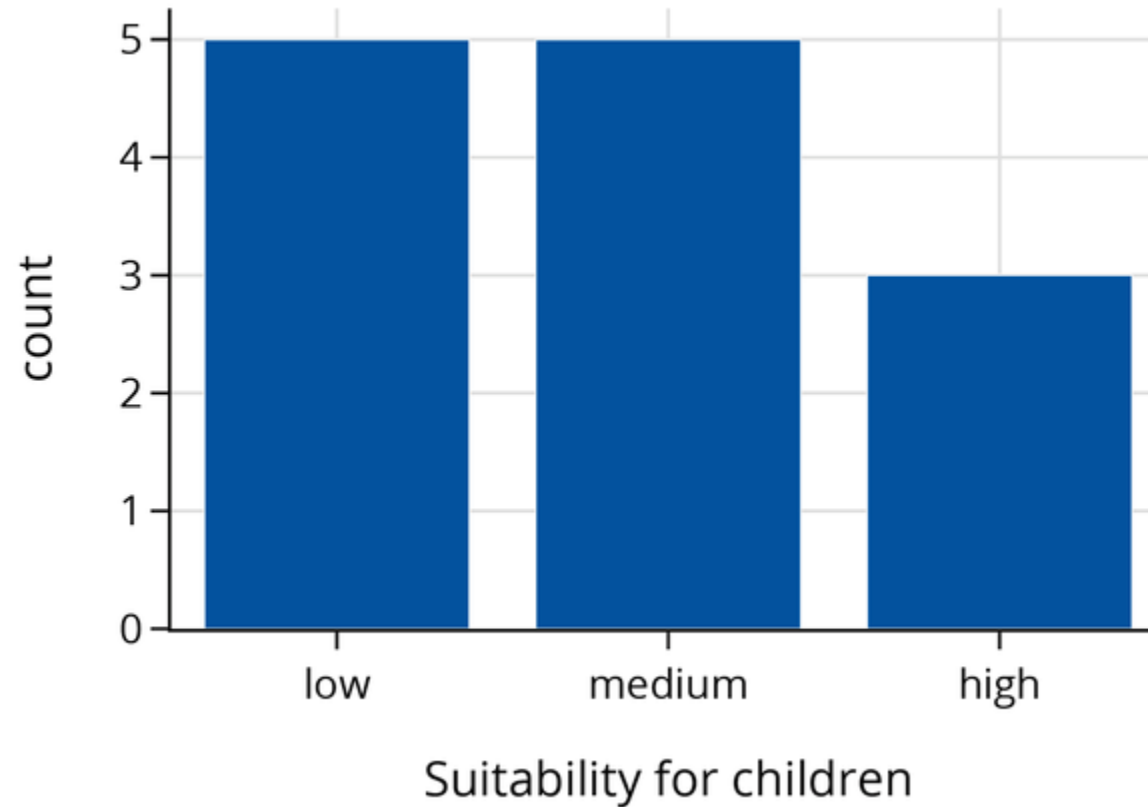
Categorical
Count and proportion



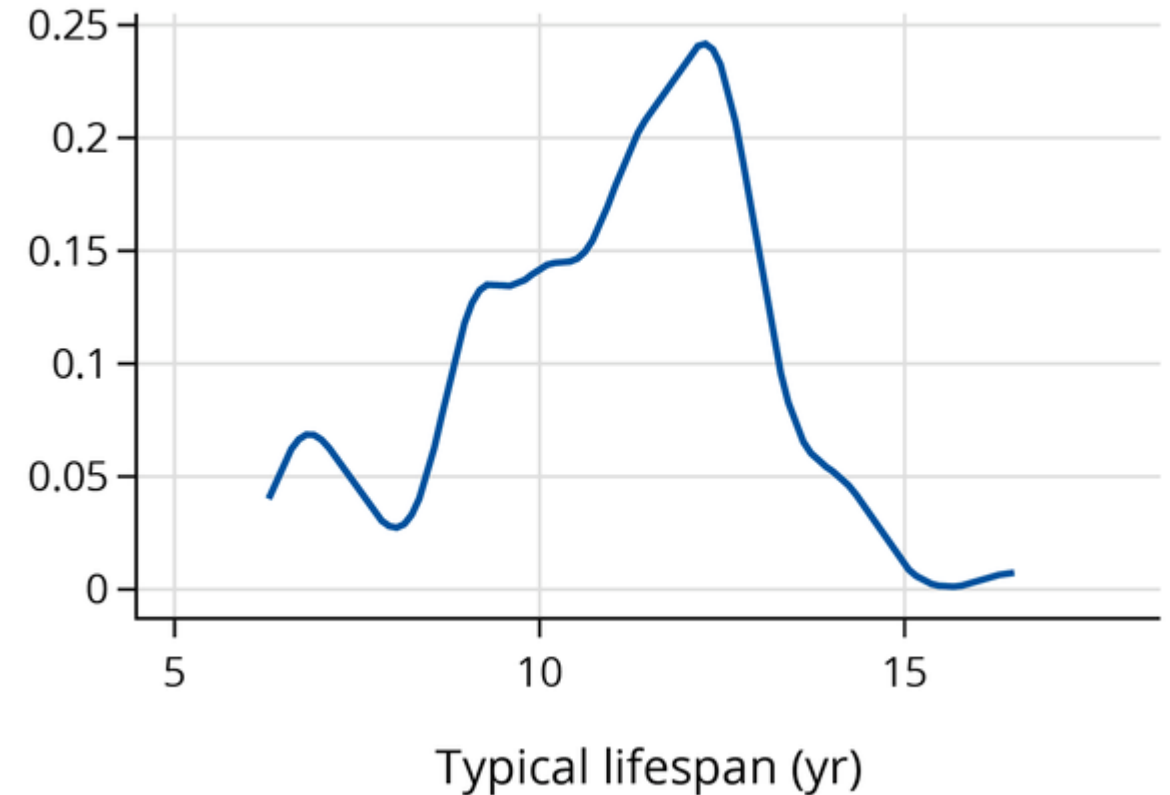
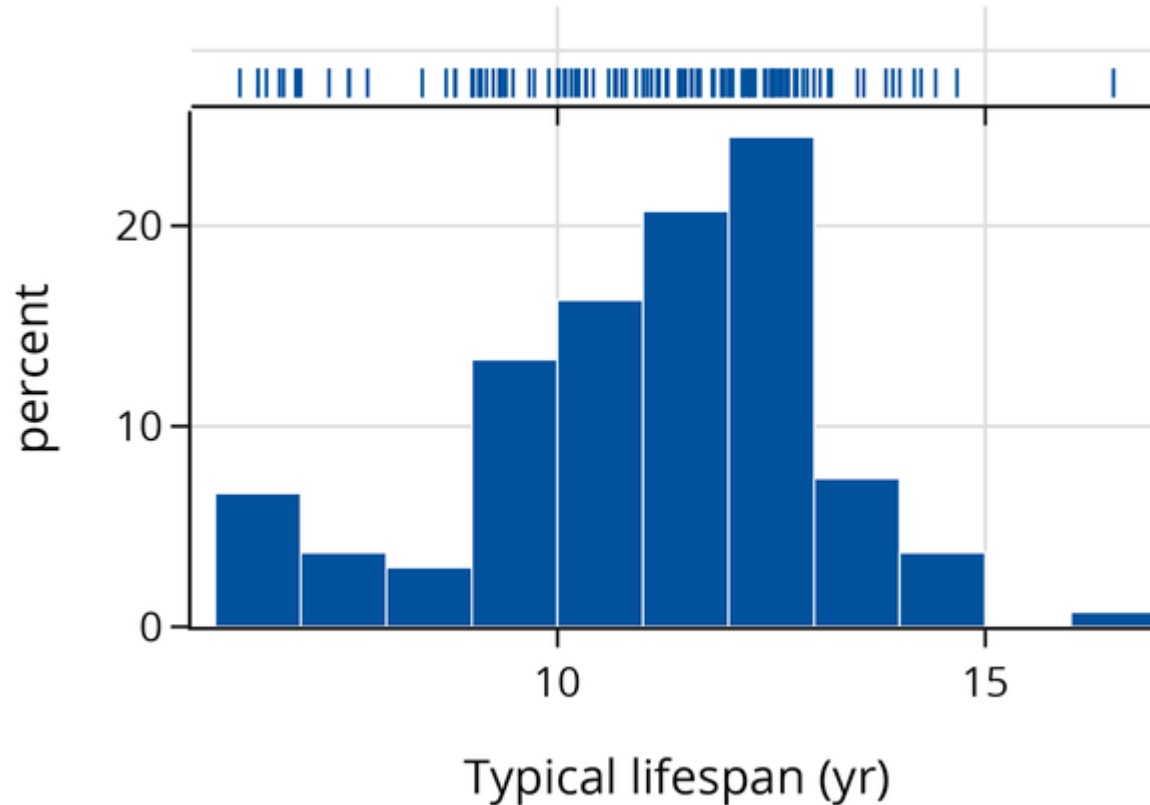
Numerical
Center, spread, shape



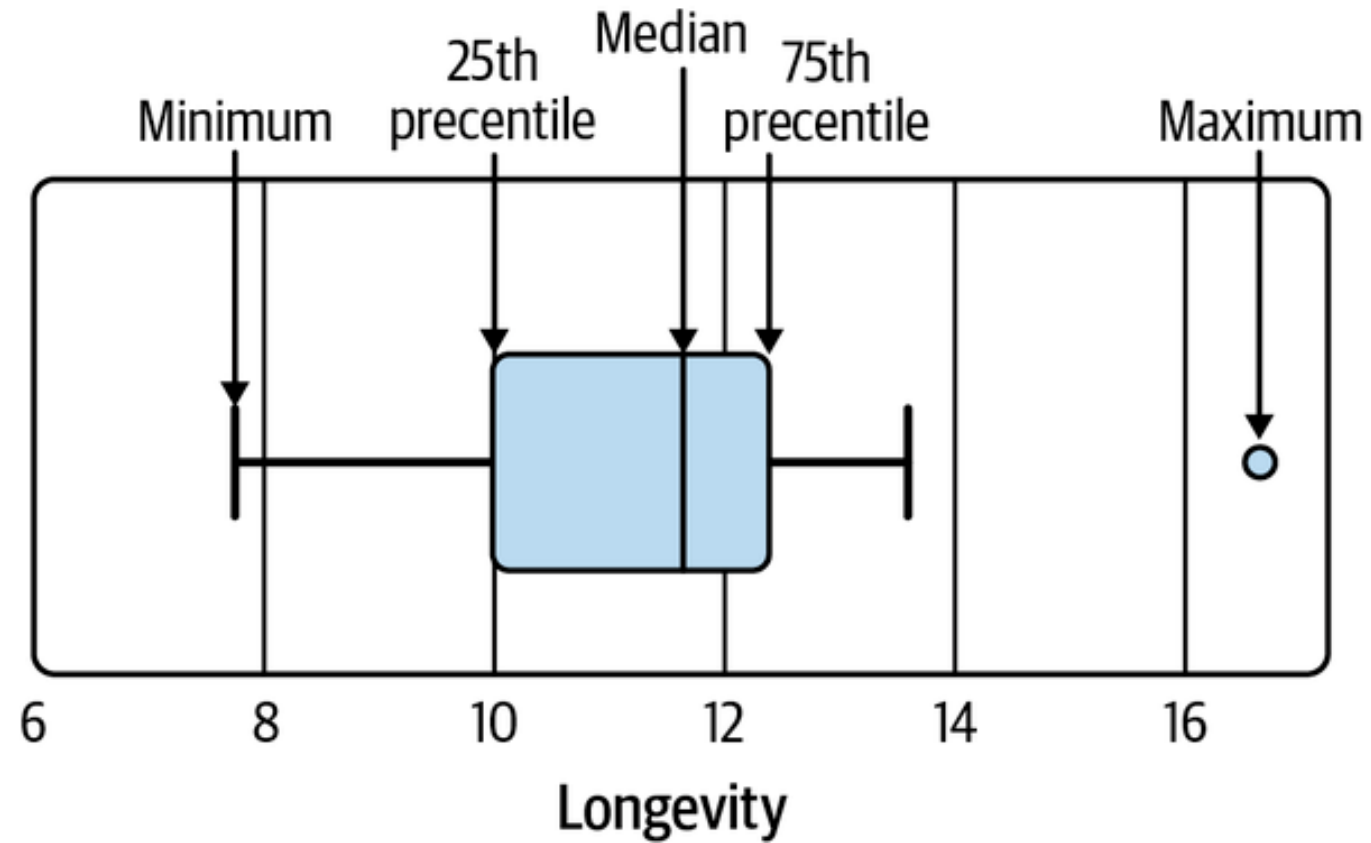
Frequency Chart



Histogram / Distribution



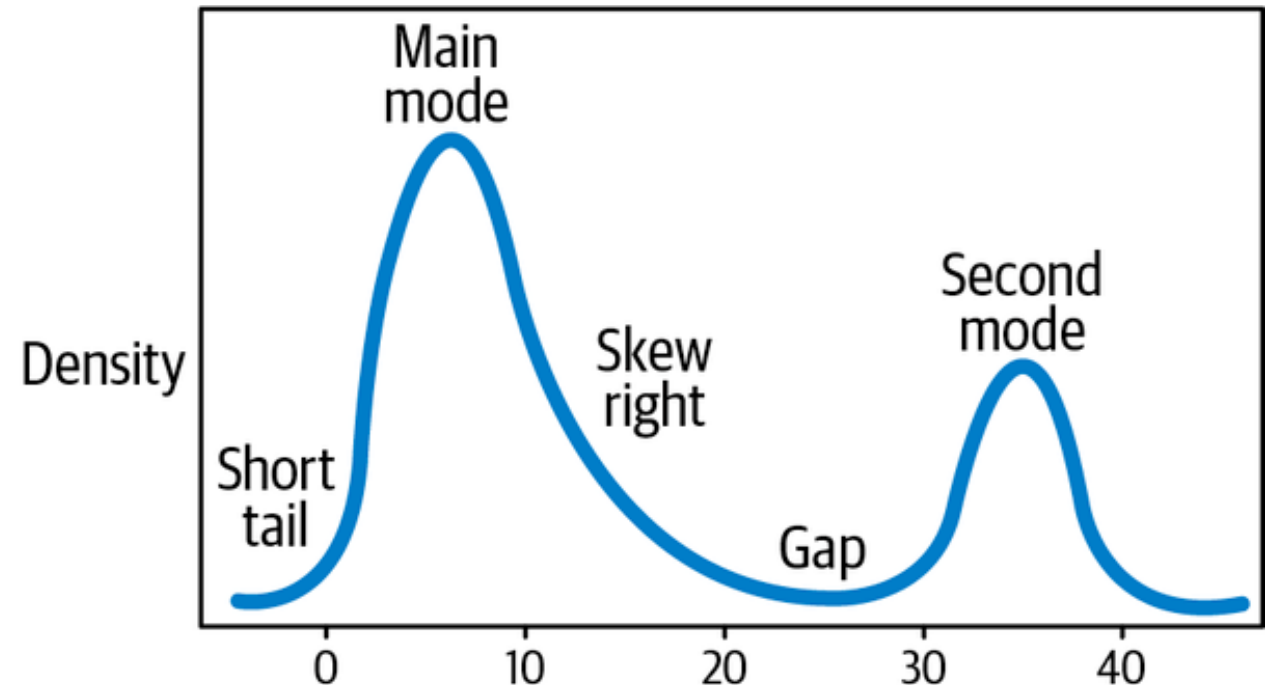
Box Plot



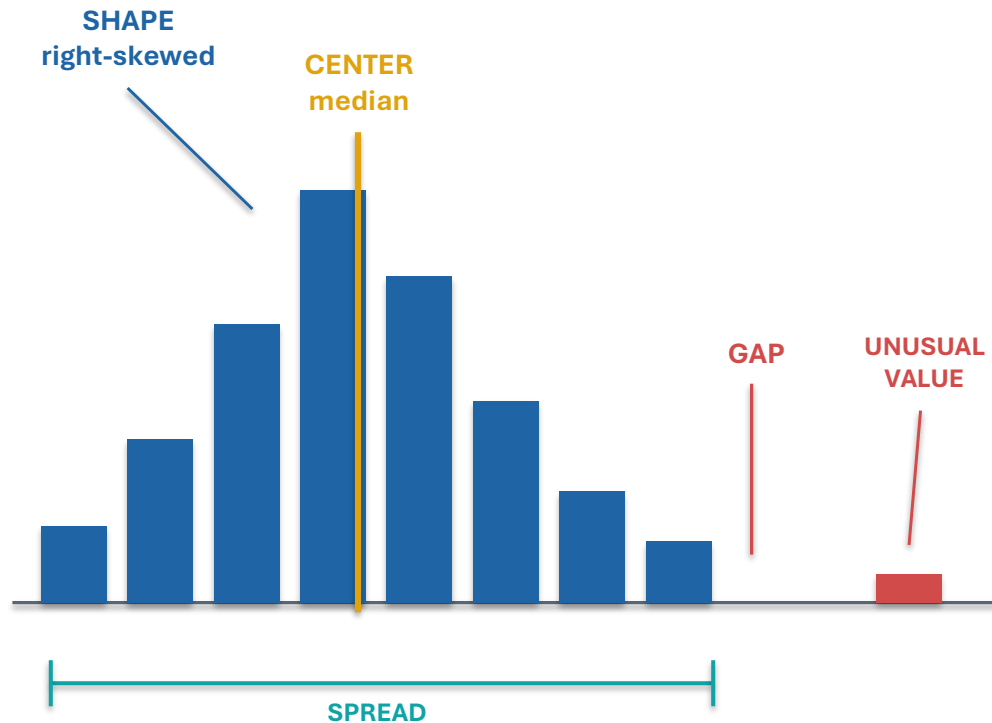
What to Look for in Distribution

We examine:

- The symmetry and skewness of the distribution
- The number, location, and size of high-frequency regions (modes);
- The length of tails (often in comparison to a bell-shaped curve)
- The gaps where no values are observed
- Anomalous values



Reading a Numerical Distribution



Shape

symmetry, skewness, modes

Center

typical value / median

Spread

range and variability

Gaps

areas with no observations

Unusual

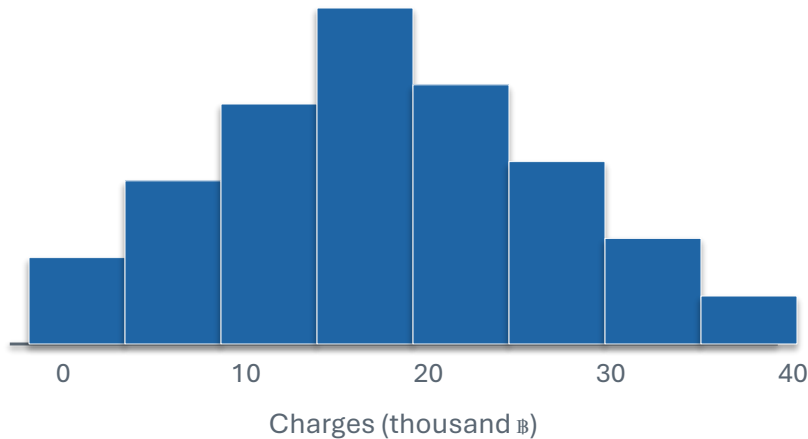
possible errors or rare cases

Histogram ≠ Bar Plot

Histogram

Numerical variable · continuous scale

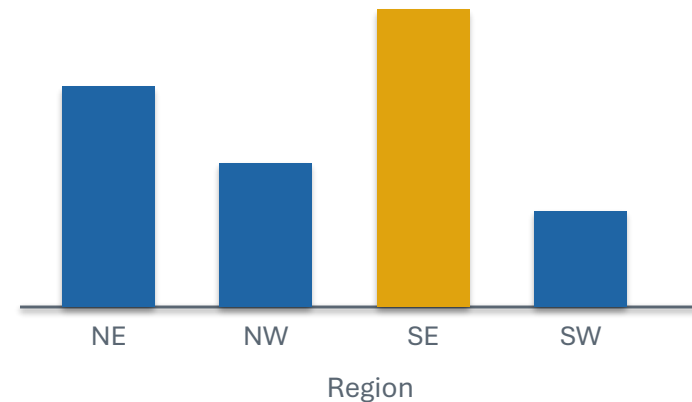
Bins represent adjacent numeric ranges → bars touch



Bar plot

Categorical variable · named groups

Categories are separate → order/spacing is not numeric distance



Do not describe a categorical bar chart as “skewed” or “long-tailed.”

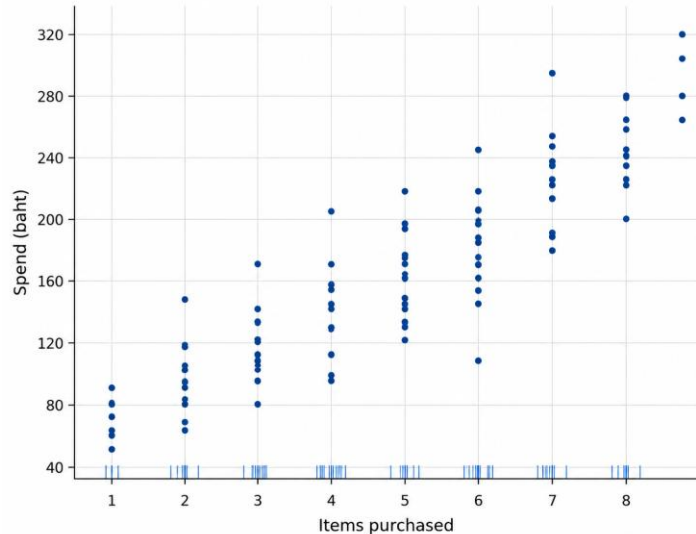
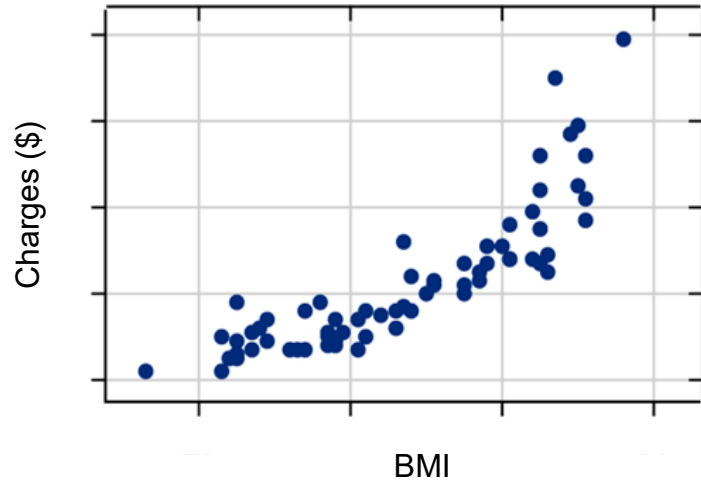


Multivariate Data Analysis

Multivariate Data Analysis

- Two numerical features
- One numerical and one categorical features
- Two categorical features
- Multivariate

Bivariate EDA: Two Numerical Features



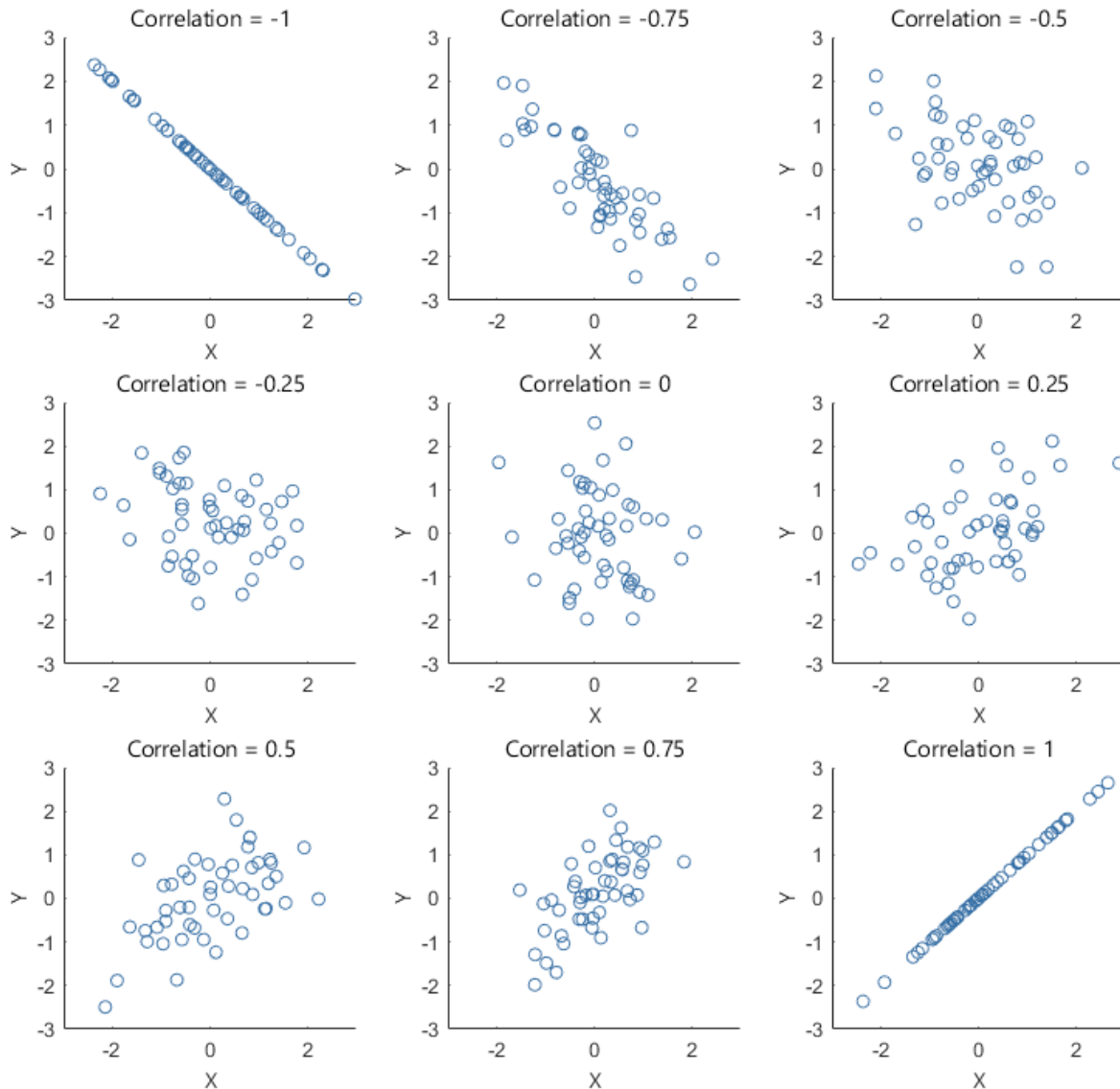
Look for:

- **Direction:** positive / negative / none
- **Form:** linear or nonlinear
- **Strength:** strong or weak pattern
- **Unusual** observation or subgroups

Example question

Do customers who buy more items tend to spend more?

Realizations of couples of random variables X and Y with different correlation coefficients



Correlation: Useful, but Easy to Misread

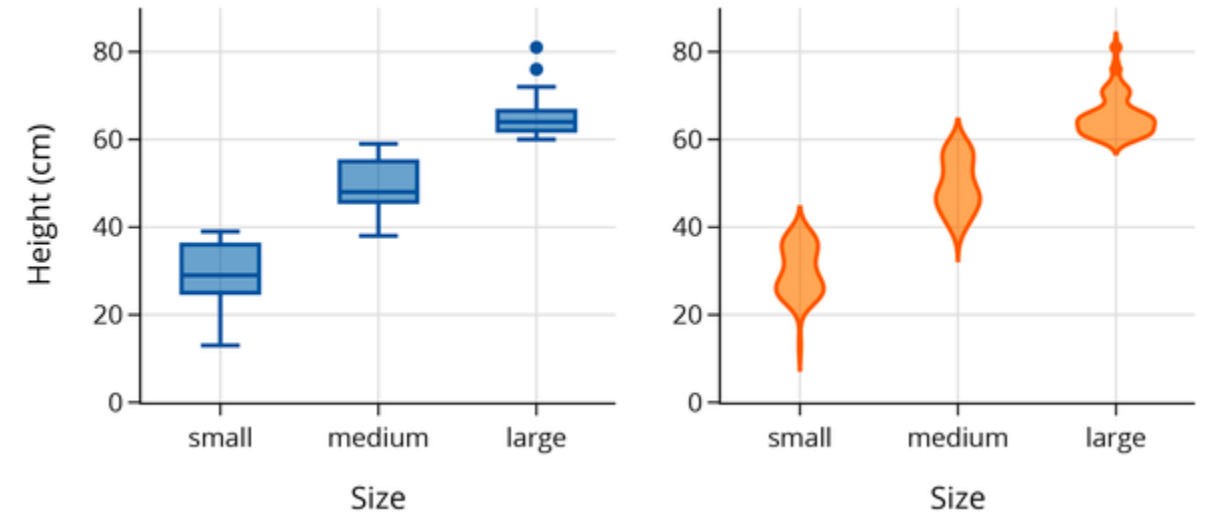
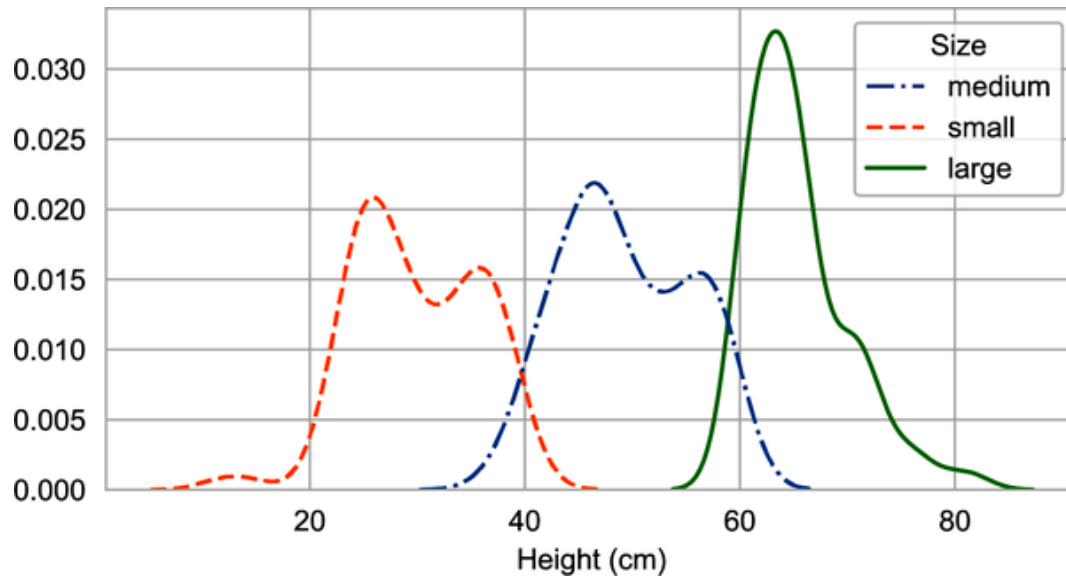
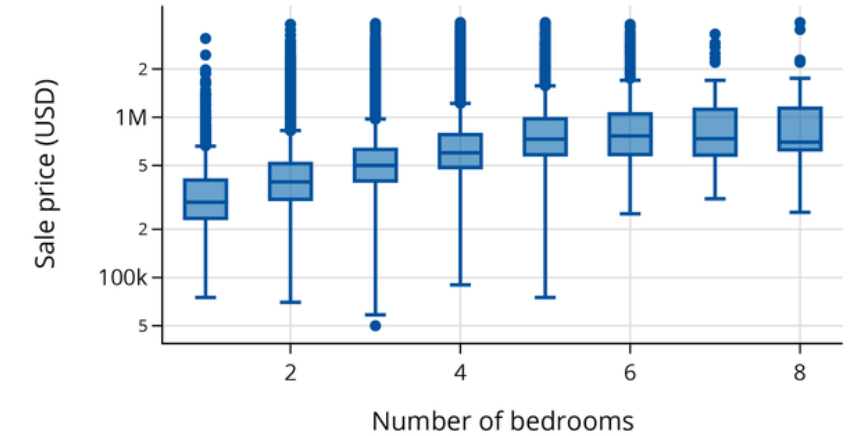
- Correlation summarizes linear association between numerical variables.
- Correlation does not prove causation.
- Low correlation does not always mean “no relationship.” Check the scatterplot.



```
corr = df.corr()
sns.heatmap(corr, annot=True)
```

One numerical and one categorical features

- Use the categorical feature to divide the data into groups and compare the distribution of the numerical feature across these groups.
- Guiding question: Do height differ between small, medium, and large size?**



One numerical and one categorical features: Compare Distributions Across Groups

Guiding question: Do waiting times differ between Counter and Mobile orders?

Lecture 6 → You already know how to produce the grouped table.

```
df.groupby('channel')['wait_time_min']  
  .agg(['count', 'mean', 'median'])
```

group	count	mean	median
Counter	163	10.2	10.1
Mobile	77	6.4	6.3

same comparison



Plot = visual form of the same grouped comparison

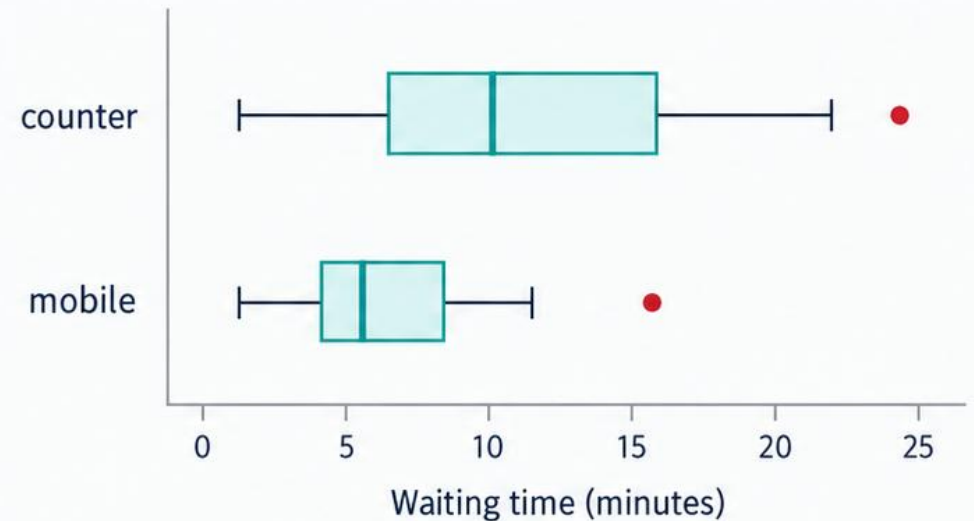
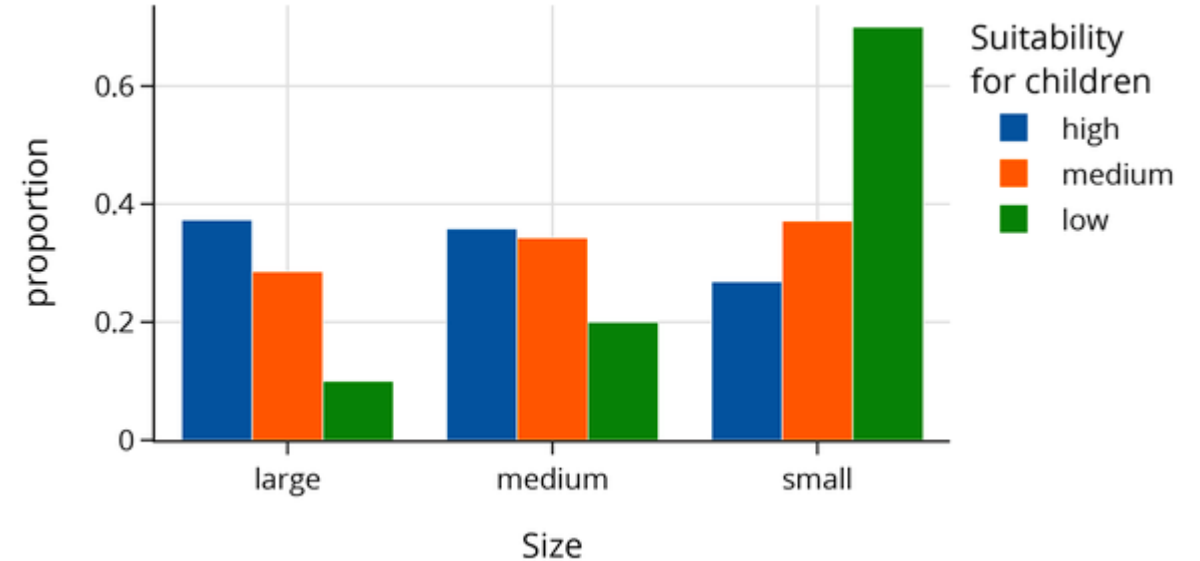
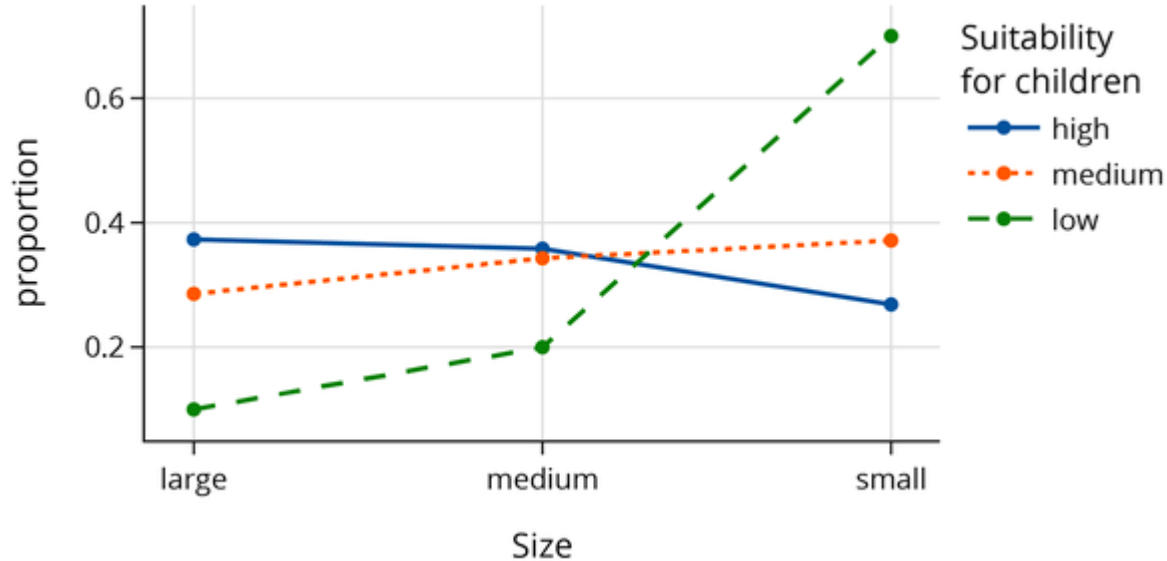


Table gives exact magnitude; plot reveals spread, overlap, and unusual values.

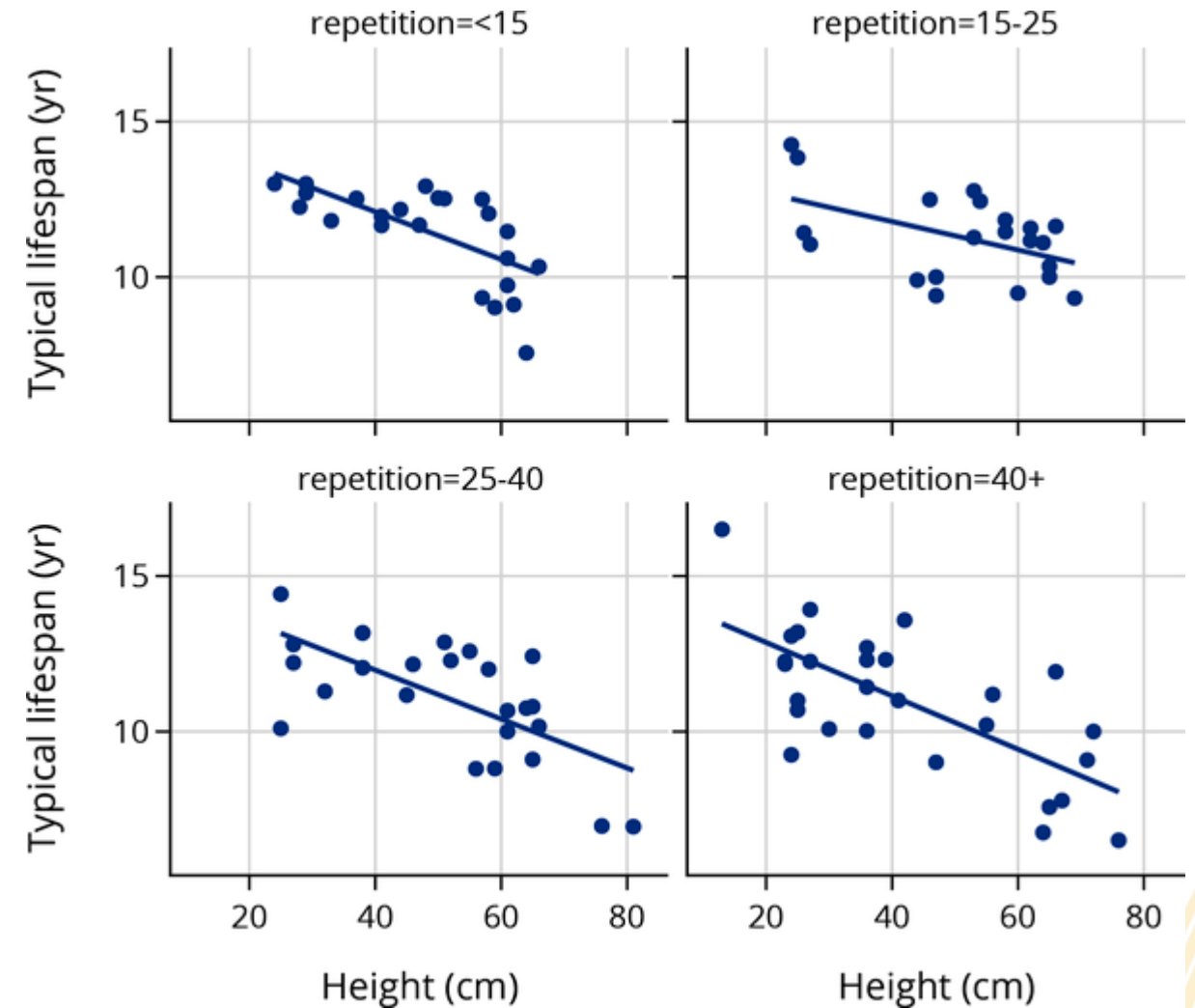
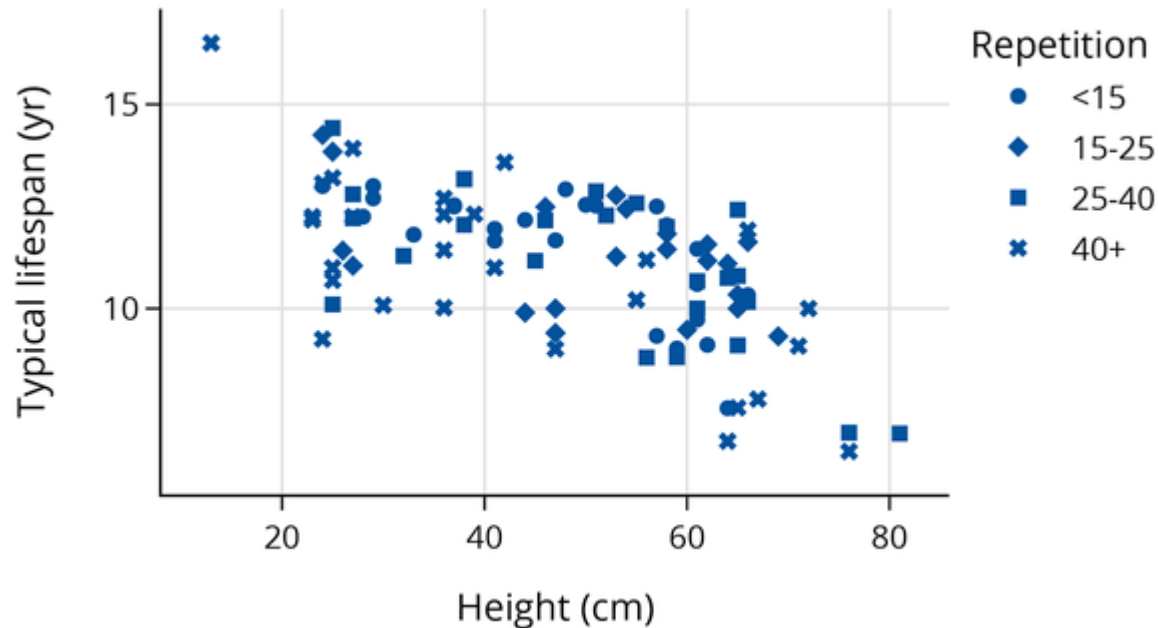
Two categorical features: Counts or Proportions

- Compare the distribution of one feature across subgroups defined by the other feature



Multivariate View

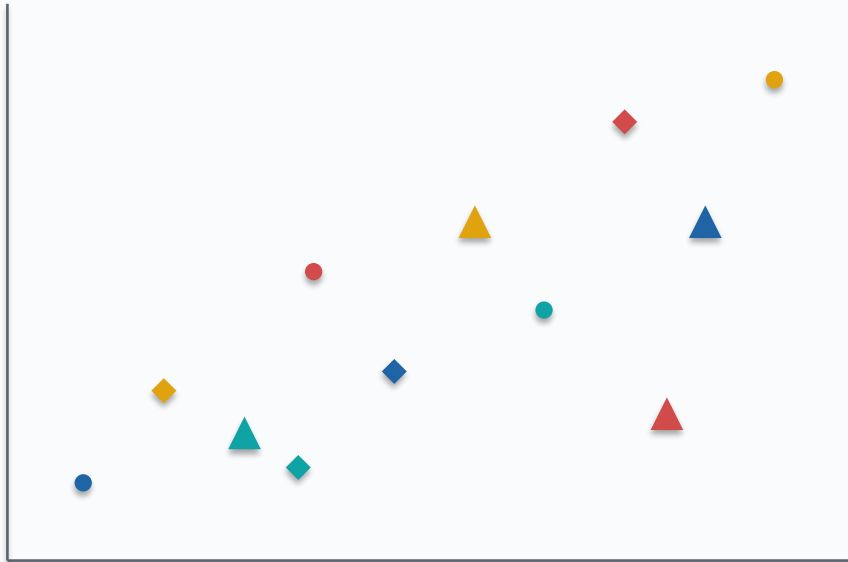
- Same techniques above with more variations in marker sizes, styles, colors, etc.



Multivariate Views: Add Complexity Deliberately

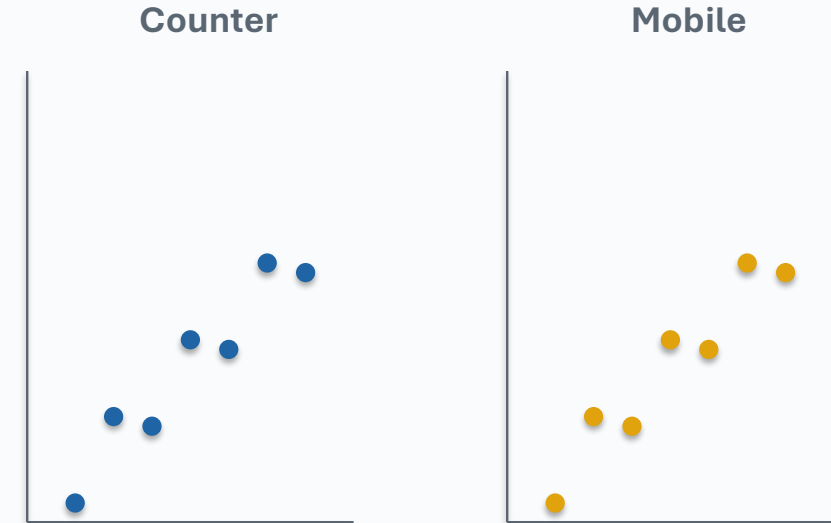
Same question, different design choices

Cluttered: every visual channel at once



hue + style + size + labels + one panel

Cleaner: add one grouping; facet when needed



Each added visual channel should answer a specific question.

Start with 2 variables → add 1 grouping → facet only when the comparison becomes clearer.

Dimensionality Reduction

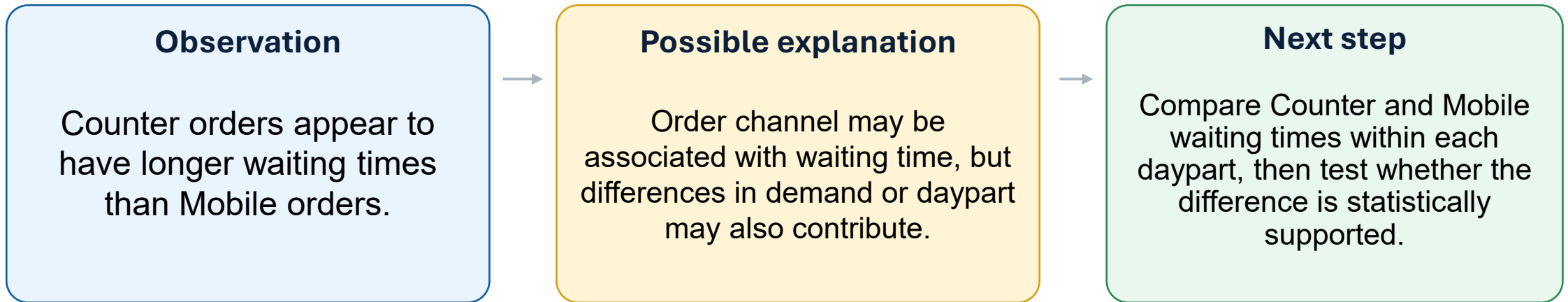
If there are too many features (or columns), we can filter out some highly-correlated features, and low-variance features.

- Variance threshold
- Univariate feature selection



Hypothesis / Insight

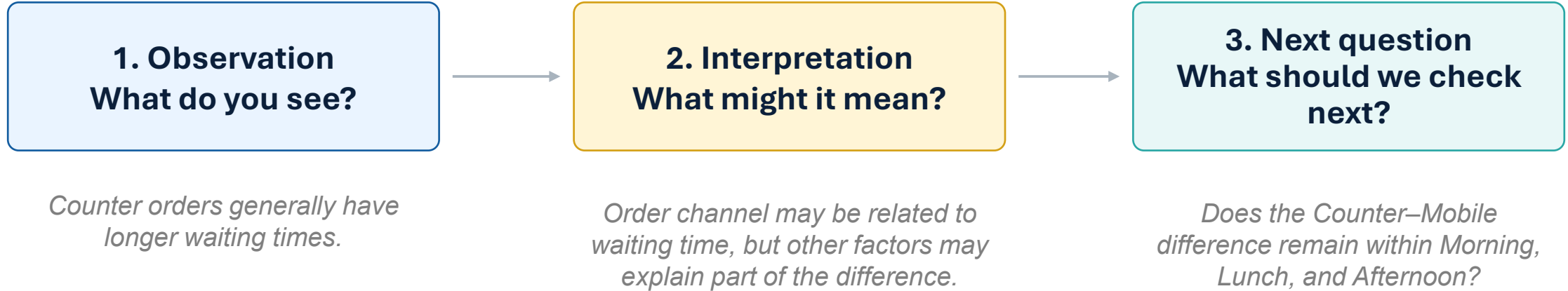
EDA Generates Hypotheses; It Does Not Prove Them



Safe wording: “associated with,” “appears higher/lower,” “worth investigating.”

Avoid overclaiming: “causes,” “proves,” or “the main reason” from EDA alone.

From Plot to Insight



Example

Counter orders have longer waiting times than Mobile orders, although the distributions overlap. The difference may be related to order channel, but lunch-period demand could also contribute.

Next, compare waiting times by channel within each daypart.

Visualization Quality: EDA vs Communication

EDA plot
Fast, flexible, for thinking

Communication plot
Clear, labeled, designed for others

Minimum quality checklist

- ✓ Meaningful title
- ✓ Labeled axes and units
- ✓ Readable categories
- ✓ Color represents information
- ✓ Avoid misleading scales

Today: mostly EDA.

Later: storytelling and dashboards.

Key Takeaways

- 1 EDA is an iterative thinking process, not a checklist of plots.
- 2 Feature type and question determine the right summary or visualization.
- 3 Use summaries and plots together: tables give magnitude, plots reveal shape.
- 4 EDA can reveal associations and hypotheses, but it does not prove causation or predictive importance.
- 5 A good EDA output ends with an observation, interpretation, and next question.



Lab Exercises

Lab Context: Insurance Claim EDA

Business situation

The insurer wants to understand claim patterns before building a model.

EDA goal

Identify patterns, differences, and unusual values worth investigating.

Guiding questions

- How are policyholders distributed across characteristics?
- Which groups have noticeably different claim rates or charges?
- Which numerical variables appear related?
- What cannot be concluded from EDA alone?

Lab Flow

1. Load & inspect

2. Classify feature types

3. Quick readiness check

4. Univariate exploration

5. Bivariate exploration

6. EDA investigation

Expected output: selected plots + short written interpretations, not just code cells.